SkillUp

Lab: Systems Testing and Deployment with Generative AI

Estimated time: 30 minutes

Learning objectives

After completing this lab, you will be able to:

- Generate test cases from requirements using generative AI
- Create deployment and rollback plans using generative AI
- Develop user acceptance testing steps using generative AI

Introduction

This activity guides you through using generative AI to support software delivery tasks. You will generate test cases, user acceptance testing (UAT) steps, deployment checklists, and rollback plans based on a set of requirements. These skills help you move from requirements to action plans quickly and with fewer mistakes. By the end, you will have practiced using AI prompts to create essential documents that support a complete delivery workflow.

Note: Make sure you have access to an AI-powered text generation tool before starting.

Exercise 1: Generate test cases and UAT steps using AI

In this exercise, you will use generative AI to turn software requirements into test cases and user acceptance testing steps. You will work with a sample requirements document and prompt the AI to create comprehensive testing plans that cover different user actions and system behaviors.

Step 1: Generate test cases from requirements

Open your AI-powered text generation tool.

Provide the following sample requirements to the AI:

```
The online shopping cart system should allow users to add items to their cart, view cart contents, update item quantities, remove items, and proceed to checkout. Additionally, the system should be able to calculate the total price, including taxes and shipping. Finally, users should be able to save their cart for later and retrieve it on their next visit.
```

Prompt the AI tool to generate test cases based on your understanding.

Review the generated test cases from the AI.

Hint

Sample prompt:

```
"Generate a comprehensive list of test cases that cover all user actions and system behaviors described in these requirements."
```

Potential AI output:

1. Verify if the user can add a single item to the cart.
2. Verify if the user can add multiple different items to the cart.
3. Verify if the user can view all items in their cart with correct details.
4. Verify if the user can update the quantity of an item in the cart.
5. Verify if the user can remove an item from the cart.
6. Verify if the cart calculates the correct subtotal for all items.
7. Verify if the cart calculates correct taxes based on the shipping location.
8. Verify if the cart calculates correct shipping costs.
9. Verify if the cart displays the correct total price (subtotal + taxes + shipping).
10. Verify if the user can save the cart for later retrieval.
11. Verify if the saved cart persists when the user logs out and logs back in.
12. Verify if the user can proceed to checkout with items in the cart.

Step 2: Validate test cases for completeness

Review the test cases generated by the AI.

Prompt the AI tool to identify any gaps based on your understanding.
Review AI's suggestions for additional test cases.

Hint

Sample prompt:

```
"Review these test cases and identify any missing scenarios or edge cases that should be tested."
```

Potential AI output:

- Test adding an item with quantity of zero.
- Test adding an item that is out of stock.
- Test updating cart quantity to exceed available inventory.
- Test removing all items from the cart (empty cart scenario).
- Test cart behavior when user is not logged in.
- Test cart behavior when session expires.
- Test proceeding to checkout with an empty cart.
- Test cart persistence across different devices for the same user.

Step 3: Generate UAT steps from requirements

Using the same requirements from Step 1, prompt the AI tool to generate UAT steps based on your understanding.

Review the generated UAT steps from the AI.

Hint

Sample prompt:

```
"Create user acceptance testing steps that validate all user stories in these requirements from an end-user perspective."
```

Potential AI output:

1. After the user browses products and adds multiple items to the cart, verify if all the items appear correctly.
2. User views cart and confirms all item details match selections, including name, price, and quantity.
3. After the user updates the quantity of an item, verify the price updates accordingly.
4. When the user removes an item from the cart, verify if the item is removed and the total is recalculated.
5. User verifies tax calculation based on shipping address.
6. User verifies shipping cost calculation based on selected shipping method.
7. User reviews final total and confirms it matches expectations.
8. After the user saves the cart and logs out, verify if the cart persists.
9. When the user logs back in, verify if the saved cart is restored with all items.
10. User proceeds to checkout successfully.

Exercise 2: Create deployment and rollback plans using AI

In this exercise, you will use generative AI to create a deployment checklist and rollback plan for your software. You will start with a set of requirements and use AI prompts to guide the creation of step-by-step plans for deploying the system and recovering if something goes wrong.

Step 1: Generate a deployment checklist

Open your AI-powered text generation tool.

Provide the following deployment context to the AI:

```
The online shopping cart system needs to be deployed to a production environment. The system includes a web application, database, and payment processing integration. The deployment must minimize downtime and ensure data integrity.
```

Prompt the AI tool to generate a deployment checklist based on your understanding.
Review the generated deployment checklist from the AI.

Hint

Sample prompt:

```
"Create a detailed deployment checklist for this system, including all steps needed to prepare, deploy, and verify the system."
```

Potential AI output:

Pre-Deployment:

- Back up current production database.
- Verify all tests pass in staging environment.
- Notify users of scheduled maintenance window.
- Create deployment rollback plan.
- Verify all environment variables are configured correctly.

Deployment:

- Enable maintenance mode on the production site.
- Deploy database schema changes.
- Deploy application code to the production servers.
- Deploy updated configuration files.
- Restart the application services.

Post-Deployment:

- Verify that the database connections are working fine.
- Run smoke tests on critical features such as login, cart, and checkout.
- Verify if the payment processing integration is functional.
- Monitor system logs for errors.
- Disable maintenance mode.
- Monitor system performance for 30 minutes.
- Notify users that system is back online.

Step 2: Create a rollback plan

Using the same deployment context from Step 1, prompt the AI tool to create a rollback plan based on your understanding.
Review the generated rollback plan from the AI.

Hint

Sample prompt:

```
"Create a rollback plan for this deployment in case of critical failures, including steps to restore the previous working version."
```

Potential AI output:

1. Immediately enable maintenance mode to prevent new transactions.
2. Stop all application services on production servers.
3. Restore database from pre-deployment backup.
4. Verify database restore completed successfully.
5. Deploy previous application code version to production servers.
6. Restore previous configuration files.
7. Restart the application services.
8. Verify if the database connections are working.
9. Run smoke tests on critical features.
10. Verify if the payment processing integration is functional.
11. Monitor system logs for errors.
12. If all tests pass, disable maintenance mode.
13. Notify users that the system is restored.
14. Document the failure and rollback for post-mortem analysis.

Step 3: Validate deployment and rollback plans

Review both the deployment checklist and rollback plan generated by the AI.
Prompt the AI tool to identify any missing steps based on your understanding.
Review the AI's suggestions for improvements.

Hint

Sample prompt:

```
"Review these deployment and rollback plans and identify any missing steps or potential risks that should be addressed."
```

Potential AI output:

- Consider adding a step to verify SSL certificates are valid before deployment.
- Add monitoring alert thresholds to detect issues quickly during deployment.
- Include a communication plan for notifying stakeholders of deployment status.
- Add a step to verify third-party integrations (payment gateway, shipping APIs) are functioning.
- Include estimated time durations for each deployment phase.
- Add a contingency plan if rollback also fails.
- Consider implementing a canary deployment or blue-green deployment strategy for lower risk.
- Add a step to review and clear any cached data that might cause issues.

Summary

You used generative AI to generate test cases, user acceptance testing steps, deployment checklists, and rollback procedures from requirements. These outputs support a smoother and more reliable software delivery process.